Q-SHIFT: An Adaptive Quantum Error-Correction Framework for Nonstationary Noise

RESEARCH SUMMARY | SIMULATION-BASED PROTOTYPE

STIM CIRCUIT Simulated shots	DETECTOR SIGNALS Recent blocks	DECODER BANK Choose next model	PAIRED CHECK Logical errors

PROBLEM AND OBJECTIVE

Quantum computers store information in delicate physical states. Small faults can alter a calculation, and fault types can change as a device operates. A logical error changes the protected information. A decoder uses measured error signals to infer and correct likely faults. A fixed decoder keeps one model throughout, which may be less suitable when noise changes. Q-SHIFT asks whether detector measurements can reveal a change early enough to select a better decoder. Its hypothesis is that causal selection may reduce logical errors compared with a strong fixed method.

METHOD AND PROTOTYPE

Using Python, Stim simulates a distance-3 rotated surface-code memory for five rounds; PyMatching decodes its detector signals. The prototype varies gate and measurement noise and uses nine models with different assumed noise strengths. It summarizes detector activity from completed blocks and may select a model for the next block, with a fixed-method fallback. All methods are compared on the same shots and observable outcomes, enabling paired comparisons. The workflow records random seeds, model choices, errors, and uncertainty.

EVALUATION AND FINDINGS

Early small-code trials showed that recognizing a noise regime does not itself improve correction; guarded selection often stayed with fixed matching. In 600,000 measurement-heavy shots, Phase 4A calibration reduced residual errors from 1,118 to 940. Phase 4B repeated the condition with 1,165 fixed errors and 1,008 calibrated errors. These support a condition-specific static-calibration effect against the nominal baseline, not adaptive superiority. A broader Phase 4C test compared calibration with stronger pooled fixed decoding across three regimes and 1.8 million paired shots. The equal-regime difference (calibrated minus pooled) was +0.0000156 (95% CI -0.0000489 to +0.0000767; p=0.667), giving no evidence of an overall gain. Short adaptive development schedules also showed no reliable advantage.

CONTRIBUTION, LIMITS, AND CONCLUSION

Q-SHIFT provides a reproducible way to test decoder specialization and adaptation using paired comparisons. Results show that estimating noise is not enough: decoder choices must deliver stable practical gains. Evidence is limited to simulation, one small code distance, selected Pauli-noise models, and short exploratory adaptive schedules. Stage 2 test predictions were independently re-decoded, but calibration samples were not retained, so the fitting step cannot be fully reproduced. No hardware integration or reliable adaptive improvement has been demonstrated. The prototype is a useful platform; the next step is to improve the noise and decoder models, then test a frozen policy on fresh data.

EVIDENCE NOTE Stage 2 test predictions were independently re-decoded, but calibration samples were not saved. Adaptive results remain exploratory; no hardware integration or confirmatory adaptive gain is claimed.